Department of AI_DS
Academic Year: 2026-27 (Odd Sem)
Course: Python for Data Science Lab (DJS23SLPC303)
Second Year B. Tech Semester: III

PRACTICAL 1

Part A (To be referred by students)

Data Types, Input / Output Statements, Operators (Arithmetic, Relational, Logical, Bitwise) and Expressions

Aim: To implement program to demonstrate data types operator and input output in python.

Problem Statements:

1. Create a program that takes user input and determines its data type.
2. Write a program to enter a number and print its square, cube and square root.
3. Write a program to perform arithmetic operations on two numbers.
4. Create a program that uses all comparison operators and prints the results.
5. Write a program to calculate the area of a right-angle triangle.
6. Calculate average of 3 numbers.
7. Calculate simple interest.
8. Convert Celsius to Fahrenheit.
9. Find Euclidean distance between two points.
10. Swap two variables using: Temp variable, Comma operator, Bitwise XOR
11. Find roots of a quadratic equation $ax^2 + bx + c = 0$.

Learning Objective: The Learner will be able to

1. Analyze the scenario to write a script declaring variables, printing output and using expressions.
2. Infer the features of various operators for different operations as per variable data type.
3. Evaluate the working of the algorithm using variables, printing and inputting user values and expressions.

Theory:

- **Data types:**
  Number: int, float, complex
  String

```
a = 11.80
x  = 100
y = "Hello"
z = 3+5.j

print(a)
print(x)
print(y)
print(z)
```

Output:

```
11.8
100
Hello
(3+5j)
```

- **Accepting Input from the Console:**

```
# if input() used it accepts input as string
num=input('Enter a number: ')
print(num)

# to accept numbers
num=int(input('Enter a number: '))
print(num)

num=float(input('Enter a number: '))
print(num)
```

Output:
Enter a number: 10
'10'
Enter a number: 45
45
Enter a number: 78
78.0

- **Output using print() with formatting:**

```
x=6
y=89
print('The value of x is {1} and y is {0}'.format(x,y))
x = 12.3456789
print('The value of x is %3.2f' %x)

print('The value of x is %3.4f' %x)

print ('The %.2f and %.1f ' %(x, y))
```

Output:
The value of x is 89 and y is 6
The value of x is 12.35
The value of x is 12.3457
The 12.35 and 456.8

```
name = "Alice"
age = 25
print(f"{name} is {age} years old.")
```

Output:
Alice is 25 years old.

- **Operators:**

Operators are used to perform operations on variables and values.
Python divides the operators in the following groups:

- **Arithmetic operators**
- Assignment operators
- **Relational/Comparison operators**
- **Logical operators**
- Identity operators
- Membership operators
- **Bitwise operators**

- **Python Arithmetic Operators**

Arithmetic operators are used with numeric values to perform common mathematical operations:

Operator	Example
+	**x + y**
-	**x - y**
*	**x * y**
/	**x / y**
%	**x % y**
**	**x ** y**
//	x // y

- **Python Assignment Operators**

Assignment operators are used to assign values to variables:

Operator	Example

=	x = 5
+=	x += 3
-=	x -= 3
*=	x *= 3
/=	x /= 3
%=	x %= 3
//=	x //= 3
**=	x **= 3
&=	x &= 3
|=	x |= 3
^=	x ^= 3
>>=	x >>= 3
<<=	x <<= 3

- **Python Relational/Comparison Operators**

Comparison operators are used to compare two values:

Operator	Example
==	x == y
!=	x != y
>	x > y
<	x < y
>=	x >= y
<=	x <= y

- **Python Logical Operators**

Logical operators are used to combine conditional statements:

Operator	Description	Example
and	Returns True if both statements are true	x < 5 and x < 10
or	Returns True if one of the statements is true	x < 5 or x < 4
not	Reverse the result, returns False if the result is true	not(x < 5 and x < 10)

- **Python Bitwise Operators**

Bitwise operators are used to compare (binary) numbers:

Operator	Name	Description
&	AND	Sets each bit to 1 if both bits are 1
|	OR	Sets each bit to 1 if one of two bits is 1
^	XOR	Sets each bit to 1 if only one of two bits is 1
~	NOT	Inverts all the bits
<<	Zero fill left shift	Shift left by pushing zeros in from the right and let the leftmost bits fall off
>>	Signed right shift	Shift right by pushing copies of the leftmost bit in from the left, and let the rightmost bits fall off

PRACTICAL 1

Part B (to be completed by students)

Data Types, Input / Output Statements, Operators (Arithmetic, Relational, Logical, Bitwise) and Expressions

1. All the students are required to perform the given tasks in Jupyter Notebook
2. Create a new notebook for each experiment. The filename should be RollNo_Name_Exp1)
3. In the first cell, the student must write his/her Name, roll no and class in the form of comments

4. Every program should be written in separate cells and in the given sequence
5. After completing the experiment, download the notebook in pdf format. The filename should be RollNo_Name_Exp1.pdf
6. Upload the pdf on the web portal (Final Submission)
7. Submit the handwritten experiment in the next lab session